JAE JEONG PARK

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EDUCATION

Oct. 2023– Present **University of California, Irvine, USA**
Doctorate of Science, Electrical Engineering (GPA 3.93/4.0)

Sep. 2014– Dec. 2015 **University at Buffalo, The State University of New York, USA**
Master of Science, Electrical Engineering (GPA 3.83/4.0)

Sep. 2010– Sep. 2014 **University at Buffalo, The State University of New York, USA**
Bachelor of Science, Electrical Engineering with High Distinction
Magna Cum Laude Degree Honors (GPA 3.61/4.0)

WORK EXPERIENCE

June 2022 – Mar 2023

PABLO AIR Co.,Ltd., Republic of Korea
Research Engineer (Middle Researcher)
TEAM PICASSO

- *On-Orbit Servicing for Lunar Surface Navigation (Doing)*
  - Moon Surface data from NASA rendering in Gazebo
  - Drone navigation from sensor fusion (Camera+Lidar+Radar+IMU+GNSS) using TRN (Terrain reference navigation) method

Jan 2022 – June 2022

- *Drone Maritime Autonomous Delivery Project (Drone Vision Part) (Doing)*
  - Developed ship detection system using Yolov5 and CSRT Tracker
  - Visual Odometry and navigation
  - Marker based visual precision landing using PX4
  - Simulation system using ROS 1 & Gazebo 11 (Camera Model + Virtual Delivery Circumstance)

Feb 2021 – Dec 2021

Dextech Co.,Ltd., Republic of Korea
Joint Project Executor (PABLO AIR & Dextech)

- *Inspection Drone*
  - Visual Odometry using SIFT algorithm
  - 3D plane model mesh point cloud mapping

- *Visual navigation and landing of drone in Gazebo simulation*
  - Target Ship detection among ships using Yolov5 + CNN classifier
  - Target Pixel based tracking and precision landing

Feb. 2016– Jul. 2019

HANCOM InSPACE Co., Ltd., Republic of Korea
Assistant Researcher
Division of Technological Innovation

- Maintenance of navigation robot
  - Resolved an issue where ramps (>13 deg) were mistaken for walls.
  - Securing additional safe zone for path-planning under mapped location.

- *Path Planning of racing car using MPC (Model Predictive Control) + GP (Gaussian Process)*

(S/W)
  - Designed Nominal Model of a Car Dynamics
  - Cost Optimization for optimal path-finding
  - Data based uncertainty parameter tuning using Gaussian Process

(H/W)
  - Running racecar in Gazebo simulation
  - AWS DeepRacer application of path planning

Licomm Co., Ltd., Republic of Korea
Research Engineer
Analog Department (Sep. 2017 - Jul. 2019)

- *R-ONU (RF Signal Over Glass + Optical Network Unit) Project*
  - Tested and managed R-ONU circuitry performance issues
  - Mass Production Line System maintenance of R-ONU
  - Conducted CE Certification Tests for low-cost version of R-ONU

Digital Department (Feb. 2016 - Sep. 2017)

- *SFP (Single Form-Factor Pluggable Transceiver) Project*
  - Involved in the 1.25/2.5/5/6.25Gbps transceiver circuitry design and maintenance of mass production
  - Conducted maintenance inspection tests of optic eyes and BER @ customary required temp. ranges for reliability

May. 2015– Jul. 2015

Fottu Co., Ltd., Republic of Korea
Intern (Research Assistant)

- *LD (TOSA) / PD (ROSA) / Bi-directional (BOSA) Assembly Line*
  - Conducted BOSA Temperature Stress Tests in a chamber and created the related test reports
- *Market Research for Sales*
  - Analyzed all SFP Matrix Sales companies and researched their technological specs for targeting future markets

Jun. 2011– Aug. 2011
Jun. 2014– Aug. 2014

Opticis Co., Ltd., Republic of Korea
Intern (Research Assistant)

- *Fiber Optic Extension Link Quality Testing*
  - Tested Extension compatibility of video signals to different TV vendors
  - EDID (Display Identification Data) Function Test

PROJECT EXPERIENCE

Feb. 2019– Jul. 2019 Research Engineer, LiComm Ltd.

- Project Title: ***"OBI Design"***
- Project budget: $5,000
  - Observed wavelength collisions for multiple R-ONU upstream causing OBI noise
  - Used cooled-type BOSA to design wavelengths shifting by time to remove OBI issues of R-ONU
  - Affiliated with the programming department to code an algorithm for shifting wavelengths according to scheduled timings that reduced the chances of OBI noise interfering with the diverse upstream signal

Feb. 2015– May. 2015 Researcher, Zhi Sun's Lab, University of Buffalo

- Project Title: ***"Fast Modulation using GNU Radio system"***
- Mentor : Zhi Sun (Associate Professor at the University of Buffalo)
  - Achieved the fastest rate of transmission of 120Mbps using GNU Radio with PAM, PSK, and GMSK
  - Planned research with the goal of improving limitations in communication through a Communicative Sensor network
  - Inspected channel quality and modulation methods through antennae type and direction or demodulation result comparison in order to improve insufficient transmission efficiency independent of the communication theory for which the depth of modulation must be directly proportional to the transmission rate
  - Produced a resultant increased transmission rate of arbitrarily managed binary data up

to 120Mbps through GMSK modulation

Spring 2014 Senior Design Seminar (EE 409), University of Buffalo

- Project Title: *"**Smartgrid Design**"*
  - Designed a system to calculate traffic light signal length of sensor RGB LED controlling algorithm with C++ using Tinyos, a remote Sensor
  - Collected virtual traffic volume from sensor nodes using wireless communication
  - Collected and calculated virtual traffic volume with nesC (Dialect of C lang) in order to embody a program that can play the same role as control towers that calculate the length of traffic lights depending on traffic volume
  - Calculated virtually-measured traffic volume between all sensor nodes (virtual traffic lights) at a central control facility (BaseStation) and embodied a system that increases the length of green lights at intersections with relatively higher traffic volumes

Fall 2013 RF/Microwave Circuits 1 (EE 401), University of Buffalo

- Project Title: *"**ADS Multicoupler Design**"*
  - Built 4 port multi-section coupler using ADS program
  - Optimized substrate properties, shapes, and lengths of each part of the coupler's real specs using ADS Simulation in order to satisfy RF Signal coupling of -20dB Coupling, -40 dB directivity, and 1.5dB>Insertion Loss

Spring 2013 Intro to Microproc. (CSE 380), University of Buffalo

- Project Title: *"**Game Programming**"*
  - Made a game named "Frogger", a frog crossing a road without being hit by randomly generated traffic, using ARMS Assembly Language
  - Resolved the disadvantage of the Arm Assembly of limited memory by optimizing Memory through Code modularization in order to embody a program with perfect interfacing for users and perfect logical operation for the game
  - Used GPIO controlling LED, push buttons, and 7-segment output LED to show the game's ongoing status

Fall 2012 Digital Principles (EE 278), University of Buffalo

- Project Title: *"**Door Lock**"*
  - Achieved the promising door lock security system by using logic gates
  - Produced binary converter that will convert to 4-bit binary method and Shift Register using Multisim and PSpice program in order to apply and utilize numbers of the decimal system used in Door locks for logic gates
  - Applied "Multisim" to control logic gates and change the password

CERTIFICATES & SKILLS

Computer ROS (Fluent)
Gazebo (Fluent)
C++ (Intermediate)

	Python (Fluent) Matlab (Fluent) Git (Intermediate) Keras (Intermediate) Tensorflow (Basic) ARMS Assembly Language (Intermediate) Multisim (Fluent) CAM 350 for PCB Gerber Revision (Intermediate) OrCAD for Circuitry Analysis (Basic) ADS (Advanced Design System) program Excel & PowerPoint (Fluent)
Language	English proficiency with GRE Verbal + Quantitative: 310 + Writing : 3.0
Others	Soldering for RF Tuning (3 years)

AWARDS, SCHOLARSHIPS AND HONORS

Sep. 2014	Magna Cum Laude Degree Honors
Spring 2012, Fall 2011	Dean's list, The State University of New York

VOLUNTARY/ EXTRACURRICULAR ACTIVITIES

Dec. 2021

Seminar-Learning based model prediction control, Future Launcher R&D Program Office, Korea Aerospace Research Institute (KARI)
Presenter

- Shared presentation of self-studied project (path planning of racing car)
- Shared ideas with listeners about the application of rocket landing using MPC

Aug. 2021
(3 days)

Visual SLAM Lecture, Daejeon Information & Culture Industry Promotion Agency, Daejeon, Republic of Korea
Student

- Attended lectures about "RGBD-SLAM"
- Learned basic concepts of mapping and pathfinding

Aug. 2019 –
Oct. 2019

FastCampus, Seoul, Republic of Korea
Student

- Attended lectures on Deep Learning starting with "Keras"
- Learned basic concepts of data analysis and CNN of image data